CASE 02 / COMMERCIAL WORK | IOS & ANDROID

Location & Geoalerting UgoRound

Native mobile location workflows with movement-aware checks and explicit battery tradeoffs.

iOS Development | Android App Development | Swift | Geolocation | Mobile App Development

PROBLEM

Location-aware applications for councils and emergency managers must relate a person's changing position to relevant geographic information. The engineering challenge includes background location behavior, map presentation, synchronization, and the tradeoff between frequent position checks and battery consumption. Both iOS and Android impose platform-specific constraints on that work.

CONTRIBUTION

From May 2016 to December 2021, I worked remotely with UgoRound in Australia on iOS and Android development. My scope included geoalerting, background location, map overlays, and movement-aware geofence checks. I worked with RxSwift and RxJava for reactive application flows and Couchbase Sync for data synchronization, connecting location-related behavior with the mobile application experience.

DECISIONS

The geofence approach considered movement, including speed and distance, rather than treating every location update as an identical event. That made responsiveness and battery consumption an explicit implementation tradeoff. Reactive flows supported the handling of changing application signals, while synchronization and map overlays addressed how geographic information reached the user and remained useful within the app.

RESULT

The work contributed native mobile capabilities to a location and geoalerting product intended for councils and emergency managers. It provides a concrete example of sustained remote collaboration and engineering across two mobile platforms. For a prospective client, the relevant experience is the combination of background behavior, location policies, map interfaces, and synchronization; that experience is relevant when planning features that must respond to movement while respecting operating-system limits and the practical cost of continuous background work.

ENGINEERING FLOW

Location Signals
Background location, movement, speed, and distance

→

Geofence Checks
Movement-aware checks with battery tradeoffs

→

Alerts & Sync
Geoalerting, map overlays, and Couchbase Sync

EVIDENCE

Professional experience: UgoRound, Australia | Remote | May 2016 - December 2021